

OMM–SBT: A 4D Causal-Kinetic Framework for Emergent Physical Laws

Diffuse Interaction Rings (DIR) and Emergent Matter Structures

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Abstract

We introduce the Oscillatory Mantle Model – Substantial Beating Theory (OMM–SBT), a 4D causal-kinetic framework in which physical laws emerge from the dynamics of a unified oscillatory substrate.

The model is based on the coupled evolution of an amplitude field X , a phase field Φ , and a kinetic field K , incorporating diffusion, nonlinear saturation, flux structuring, and feedback mechanisms. Within this formulation, particles, interactions, geometry, and thermodynamic behavior are not assumed as fundamental entities, but arise as emergent structures of the underlying field dynamics.

We show that the framework reproduces multiple physical sectors as effective limits, including Schrödinger-like wave behavior, Maxwell-like flux propagation, Dirac/Pauli-like vorticity structures, Einstein-like effective geometry, and Hawking-like boundary radiation. These results are supported by progressive numerical calibration, strong invariance properties, and independent reproduction of the core dynamics.

A key feature of the model is the emergence of *Diffuse Interaction Rings (DIR)*, where localized energetic stabilizations give rise to particle-like behavior. In this interpretation, matter corresponds to stable patterns within interaction structures rather than primitive objects.

The framework also introduces a generalized continuity equation with a non-zero source term, reflecting global constraints and feedback processes within the substrate. Thermodynamic behavior emerges as a global kinetic projection of the coupled sectors, reaching robust and numerically derived-candidate behavior.

While several aspects remain to be formally completed, including analytic closure, rigorous quantization, and strict Lorentz invariance, the results support the interpretation of OMM–SBT as a coherent and testable candidate for emergent physical unification.

1 Introduction

Modern theoretical physics is structured around two major frameworks: quantum field theory, which successfully describes microscopic interactions, and general relativity, which governs gravitational phenomena and spacetime geometry. Despite their respective successes, these frameworks remain formally incompatible in extreme regimes.

Numerous approaches to unification have been proposed, including higher-dimensional theories such as string theory and M-theory. These models introduce extended mathematical structures but often rely on additional spatial dimensions and remain only partially connected to experimental validation.

In this work, we propose an alternative approach based on a causal-kinetic substrate framework. Instead of introducing new dimensions or fundamental particles, we consider that physical laws emerge from a unified dynamical system defined on a finite, constrained substrate.

The core hypothesis is that physical structure is not fundamental, but emergent. Observable phenomena arise from the interaction between:

- an amplitude field X , representing the underlying field-mantle,
- a phase/orientation field Φ , encoding directional structure,
- and a kinetic field K , representing accumulated dynamical activity and feedback.

A central feature of this framework is the presence of a feedback loop in which consequences (flux, curvature, radiation) dynamically influence the substrate itself, thereby becoming renewed effective causes.

This leads to a generalized non-equilibrium structure in which local conservation laws are modified by global constraints and feedback processes.

A key structural concept introduced in this work is that of *Diffuse Interaction Rings (DIR)*, which correspond to extended interaction regions where localized energetic stabilizations give rise to particle-like behavior.

We show that multiple physical sectors—including Schrödinger-like, Dirac-like, Maxwell-like, Einstein-like, thermodynamic, Hawking-like, and extended (string-like) behaviors—emerge as limits of the same underlying dynamics.

Unlike higher-dimensional unification theories, the present framework operates entirely in an effective 3+1 dimensional setting. Higher-order structures arise as kinetic consequences rather than additional spatial dimensions.

The model is supported by numerical calibration, invariance analysis, and independent reproduction, and is presented here as a serious candidate for emergent physical unification.

The goal of this paper is to provide a complete formal description of the framework, its equations, its physical interpretations, and its testable predictions.

2 Positioning with Respect to Existing Theoretical Frameworks

The search for a unified description of physical laws has led to the development of several major theoretical frameworks. Among them, quantum field theory (QFT) and general relativity (GR) constitute the current foundations of modern physics. While QFT provides an accurate description of particle interactions and gauge symmetries, GR successfully models gravitational phenomena through spacetime curvature.

However, these frameworks remain formally incompatible in regimes where both quantum and gravitational effects are significant. This has motivated the exploration of various unification approaches.

2.1 Relation to Quantum Field Theory

Quantum field theory is built upon operator-valued fields defined over spacetime, with dynamics governed by Lagrangian or Hamiltonian formulations and quantization procedures. In contrast, the present framework does not begin with quantized operators but with a dynamical substrate described by real-valued fields (X, Φ, K) .

Quantum-like behavior emerges in this model as a consequence of:

- spectral mode structures,
- interference patterns arising from phase dynamics,
- and effective probabilistic distributions linked to the substrate's kinetic state.

Thus, rather than assuming quantization as an initial postulate, the model investigates how quantum-like behavior may arise as an emergent property of underlying field dynamics.

2.2 Relation to General Relativity

General relativity describes gravity as a geometric property of spacetime, encoded in the metric tensor $g_{\mu\nu}$. In the present framework, no fundamental spacetime geometry is assumed.

Instead, an *effective metric* g_{eff} emerges from the structure of the substrate through local quantities such as:

- density-like measures,
- gradients and Laplacians of the field X ,
- flux structures,
- and kinetic activity.

This leads to curvature-like effects that reproduce gravitational behavior as a consequence of field organization rather than as a primitive geometric postulate.

2.3 Relation to String Theory and Higher-Dimensional Models

String theory and M-theory attempt unification by introducing extended objects (strings, branes) and additional spatial dimensions. While mathematically rich, these models remain only partially connected to direct experimental validation.

In contrast, the present framework remains strictly four-dimensional (3+1). Extended structures do appear, but they are interpreted as:

- coherent low-frequency modes,
- extended dynamical patterns,
- or kinetic consequences of the substrate,

rather than fundamental higher-dimensional objects.

Thus, the model provides an alternative interpretation of string-like behavior without requiring extra spatial dimensions.

2.4 Relation to Thermodynamics and Non-Equilibrium Systems

Unlike equilibrium-based formulations, the proposed framework is intrinsically non-equilibrium. The presence of a kinetic field K and feedback mechanisms introduces:

- persistent energy circulation,
- entropy production,

- and dynamical steady states.

In this framework, thermodynamic behavior should not be interpreted as an independent sector. Instead, it emerges as a global kinetic projection of the coupled dynamics of (X, Φ, K) .

This implies that:

- temperature is linked to substrate kinetics,
- entropy reflects the distribution of kinetic activity,
- irreversibility arises from feedback and dissipation,
- and thermodynamic closure depends on the full coupled system.

This perspective aligns with modern non-equilibrium statistical physics, where macroscopic behavior arises from strongly coupled microscopic dynamics.

2.5 Conceptual Shift: From Objects to Processes

A key conceptual difference between this framework and most existing theories lies in its ontology.

Standard approaches typically assume:

- particles or fields as fundamental entities,
- and interactions as secondary processes.

In contrast, the present model adopts a process-based ontology:

- the substrate dynamics is primary,
- structures (particles, fields, geometry) are emergent,
- and physical laws correspond to stable dynamical regimes.

This shift is consistent with several modern perspectives in theoretical physics, where emergence plays a central role.

2.6 Summary of Positioning

The framework proposed here can be summarized as follows:

- It reproduces quantum-like, relativistic, and thermodynamic behaviors as emergent limits.
- It avoids introducing extra spatial dimensions.
- It incorporates feedback as a fundamental mechanism.
- It treats conservation laws as generalized constraints rather than strict local equalities.
- It introduces Diffuse Interaction Rings (DIR) as emergent interaction structures underlying matter-like localization.
- It is supported by numerical calibration and independent reproduction.

As such, it should not be viewed as a replacement of existing theories, but rather as a candidate underlying structure from which they may arise as effective descriptions.

3 Axiomatic Foundations of the OMM–SBT Framework

The Oscillatory Mantle Model – Substantial Beating Theory (OMM–SBT) is built upon a minimal set of axioms that define its ontology, dynamics, and interpretative structure. These axioms are not postulated arbitrarily, but are motivated by the requirement of reproducing known physical behaviors while maintaining conceptual coherence and minimality.

3.1 Axiom A1: Fundamental Oscillatory Excitation

The substrate is intrinsically dynamic and driven by a fundamental oscillatory process.

$$\text{Primary dynamical driver: intrinsic oscillatory excitation} \quad (1)$$

This oscillatory behavior is not externally imposed but is a defining property of the system, generating continuous fluctuations that act as the origin of all subsequent structure.

3.2 Axiom A2: Finite and Constrained Substrate

The substrate is globally constrained and effectively finite.

$$\|X\|^2 \approx 1 \quad (\text{effective global normalization}) \quad (2)$$

This constraint introduces non-trivial global effects:

- boundary-like behaviors,
- effective horizons,
- and non-local correlations.

Local dynamics must therefore be understood as embedded within a globally constrained system.

3.3 Axiom A3: Field-Mantle Ontology

The fundamental entity is not a particle but a continuous scalar field:

$$X(x, t) \in \mathbb{R} \quad (3)$$

This field represents an amplitude-like quantity forming a *mantle* from which observable structures emerge.

3.4 Axiom A4: Phase and Orientation

A phase field encodes local orientation and directional information:

$$\Phi(x, t) \in \mathbb{R} \quad (4)$$

Gradients of this phase define preferred directions and give rise to structured propagation phenomena. The phase is essential for interference, coherence, and directional coupling.

3.5 Axiom A5: Substrate Kinetics

A kinetic field $K(x, t)$ represents accumulated dynamical activity within the substrate:

$$K(x, t) \geq 0 \quad (5)$$

This quantity encodes:

- local agitation,
- memory of past dynamics,
- and energy redistribution processes.

It plays a central role in thermodynamic behavior and feedback mechanisms.

3.6 Axiom A6: Feedback of Consequences

A defining feature of the framework is the existence of a feedback loop mediated by the kinetic field K :

$$\text{Consequences} \rightarrow K \rightarrow \text{Substrate} \rightarrow \text{New consequences} \quad (6)$$

Fluxes, curvature-like effects, and boundary phenomena contribute to the evolution of K , which in turn modifies the dynamics of X and Φ .

3.7 Axiom A7: Four-Dimensional Sufficiency

The framework operates in an effective 3+1 dimensional setting:

$$(x, t) \in \mathbb{R}^3 \times \mathbb{R} \quad (7)$$

No additional spatial dimensions are introduced. Higher-order structures arise as dynamical or kinetic phenomena rather than geometric dimensions.

3.8 Axiom A8: Emergent Physical Sectors

All physical sectors are interpreted as emergent regimes of the same underlying system.

$$\text{Physical laws} = \text{stable dynamical limits of } (X, \Phi, K) \quad (8)$$

This includes:

- wave-like dynamics (Schrödinger-like),
- spinor and chirality effects (Dirac-like),
- flux-based interactions (Maxwell-like),
- curvature and geometry (Einstein-like),

- thermodynamic behavior,
- boundary radiation phenomena (Hawking-like),
- and interaction-structure-driven matter (via Diffuse Interaction Rings).

3.9 Interpretative Summary

Taken together, these axioms define a framework in which:

- the substrate dynamics is fundamental,
- observable structures are emergent,
- conservation laws are generalized,
- and causality includes feedback processes.

The remainder of this work develops the mathematical formalism and demonstrates how known physical behaviors arise from these principles.

4 Fundamental Variables and Derived Quantities

The OMM-SBT framework is defined by three primary dynamical variables:

$$\Psi_t = (X_t, \Phi_t, K_t).$$

Here, Ψ_t denotes the full state of the system at time t . The three components X_t , Φ_t , and K_t respectively describe the field-mantle amplitude, the internal phase/orientation, and the kinetic state of the substrate.

4.1 Field-Mantle Amplitude

The first fundamental variable is the field-mantle amplitude:

$$X = X(x, t).$$

The variable x denotes position in space and t denotes time. In the discrete numerical implementation, x corresponds to a lattice site and t to an iteration step.

The field X is not interpreted as a particle field in the usual sense. Rather, it represents the underlying mantle from which localized structures, effective particles, fluxes, and geometrical deformations may emerge.

A global normalization constraint is imposed:

$$\|X\|^2 \approx 1.$$

This expresses an effective global constraint on the substrate and implies that local variations of X cannot be interpreted independently of the global state of the system.

4.2 Phase and Orientation Field

The second fundamental variable is the phase field:

$$\Phi = \Phi(x, t).$$

The phase field encodes orientation, internal directionality, and coherence. Its spatial gradient defines preferred directions of propagation:

$$\nabla\Phi.$$

The phase field is therefore responsible for wave-like organization, interference-like behavior, and directional coupling.

4.3 Substrate Kinetic Field

The third fundamental variable is the kinetic field:

$$K = K(x, t), \quad K \geq 0.$$

The field K represents the local kinetic activity of the substrate. It stores the accumulated effect of fluxes, vorticity, boundary activity, and feedback. In this sense, K is not merely a temperature-like variable, but a memory-bearing kinetic reservoir.

It plays a central role in thermodynamic behavior and in the feedback loop:

$$\text{consequences} \rightarrow K \rightarrow \text{new effective causes}.$$

4.4 Effective Density

From the field-mantle and kinetic field, we define an effective density:

$$\rho_{\text{eff}} = X^2 + \alpha_K K.$$

The coefficient α_K is a dimensionless parameter controlling the contribution of substrate kinetics to the effective density.

This construction reflects the fact that density is not purely amplitude-based in OMM-SBT. Since kinetic feedback modifies the substrate, the effective density must include both field intensity and kinetic activity.

4.5 Flux Skeleton

The fundamental flux is defined by:

$$J = |X|\nabla\Phi.$$

Each symbol has a specific meaning:

- $|X|$ measures the local amplitude of the field-mantle;
- $\nabla\Phi$ measures the local phase gradient;
- J is therefore the amplitude-weighted directional flow.

This quantity is called the flux skeleton because it provides the common structure from which Maxwell-like propagation, transport, and sectoral pathways emerge.

4.6 Vorticity and Chirality

From the flux J , we define the vorticity:

$$W = \nabla \times J.$$

In two-dimensional numerical simulations, this corresponds to the scalar curl of the flux field. In three spatial dimensions, W becomes a vector field.

The vorticity measures local circulation of the flux skeleton. It is associated with:

- chiral structures,
- spinor-like behavior,
- local rotational organization,
- and Dirac/Pauli-like sector emergence.

4.7 Curvature Proxy

We define a curvature-like scalar from the Laplacian of the field:

$$C = |\nabla^2 X|.$$

This quantity should be interpreted as a scalar proxy for local curvature-like deformation of the mantle.

Regions of high C correspond to strong deviations from harmonic smoothness and contribute to effective geometry.

4.8 Effective Metric

The effective metric proxy is defined as:

$$g_{\text{eff}} = 1 + a\rho_{\text{eff}} + bC + c|W| + d|J| + eK.$$

The coefficients a, b, c, d, e determine the relative contribution of density, curvature, vorticity, flux, and kinetic activity.

This expression defines a scalar effective geometry proxy and should not be interpreted as a full relativistic metric tensor. A complete tensorial formulation remains a future analytic extension.

4.9 Effective Horizon

A horizon-like boundary is defined by thresholding the effective metric:

$$H = \{x \mid g_{\text{eff}}(x) > q_{\theta}(g_{\text{eff}})\},$$

where q_{θ} is a quantile threshold.

This construction identifies regions where the effective geometry becomes extreme relative to the global substrate. Such regions behave as finite-size causal boundaries and support Hawking-like or radiation-like behavior.

4.10 Radiation Proxy

A radiation-like field is constructed as a model-dependent combination of boundary, flux, vorticity, and curvature activity:

$$R = R(H, J, W, K, \nabla X, \nabla^2 X).$$

A typical form is:

$$R = \eta_H H + \eta_J \tanh\left(\frac{|J|}{\langle |J| \rangle}\right) + \eta_W \tanh\left(\frac{|W|}{\langle |W| \rangle}\right) + \eta_K \tanh\left(\frac{K}{\langle K \rangle}\right) + \eta_C \tanh\left(\frac{C}{\langle C \rangle}\right).$$

Here:

- H contributes boundary activity;
- J contributes flux intensity;
- W contributes rotational activity;
- K contributes substrate kinetic activity;
- C contributes mantle deformation.

The coefficients $\eta_H, \eta_J, \eta_W, \eta_K, \eta_C$ weight these contributions.

4.11 Summary of Variables

The fundamental and derived quantities are summarized as follows:

$$\Psi_t = (X_t, \Phi_t, K_t)$$

$$\rho_{\text{eff}} = X^2 + \alpha_K K$$

$$J = |X| \nabla \Phi$$

$$W = \nabla \times J$$

$$C = |\nabla^2 X|$$

$$g_{\text{eff}} = 1 + a\rho_{\text{eff}} + bC + c|W| + d|J| + eK$$

$$H = \{g_{\text{eff}} > q_\theta(g_{\text{eff}})\}$$

$$R = R(H, J, W, K, \nabla X, \nabla^2 X)$$

Together, these variables form the mathematical basis of the OMM-SBT framework. The following section introduces their time evolution and the dynamical laws governing the model.

5 Dynamical Equations of the OMM–SBT Framework

The dynamics of the OMM–SBT framework are governed by the coupled evolution of the three fundamental fields:

$$\Psi_t = (X_t, \Phi_t, K_t).$$

The evolution is discrete in time for numerical implementation, but admits a continuous interpretation in the limit $\Delta t \rightarrow 0$.

5.1 General Structure of the Dynamics

The evolution follows the general structure:

$$\Psi_{t+1} = \mathcal{F}(\Psi_t), \tag{9}$$

where $\mathcal{F}$ is a nonlinear operator encoding diffusion, dissipation, nonlinear saturation, flux interactions, and feedback.

Each component of Ψ_t evolves according to its own equation, coupled to the others.

5.2 Evolution of the Field-Mantle Amplitude X

The amplitude field evolves according to:

$$X_{t+1} = \mathcal{N} \left[X_t + D\nabla^2 X_t - \gamma X_t - \lambda X_t^3 + C_{\text{flux}} + C_{\text{finite}} + \eta_t \right]. \tag{10}$$

Each term has a precise role:

- X_t : persistence of the field from one step to the next.
- $D\nabla^2 X_t$: diffusion term, responsible for spatial propagation and smoothing.
- $-\gamma X_t$: dissipation, representing energy loss or relaxation.
- $-\lambda X_t^3$: nonlinear saturation, preventing divergence and creating attractors.
- C_{flux} : coupling to flux structures (derived from J).
- C_{finite} : global constraint term enforcing finite substrate behavior.
- η_t : fluctuation/noise term, representing intrinsic excitation.
- $\mathcal{N}[\cdot]$: normalization operator enforcing $\|X\|^2 \approx 1$.

A typical form of the finite constraint term is:

$$C_{\text{finite}} = -\mu \left(\|X_t\|^2 - 1 \right) X_t. \tag{11}$$

5.3 Evolution of the Phase Field Φ

The phase evolves according to:

$$\Phi_{t+1} = \Phi_t + \alpha \nabla^2 \Phi_t + \beta W_t + C_{\text{orient}}. \quad (12)$$

The terms are interpreted as follows:

- Φ_t : phase persistence.
- $\alpha \nabla^2 \Phi_t$: phase diffusion, promoting coherence.
- βW_t : coupling to vorticity, introducing rotational/chiral structure.
- C_{orient} : additional orientation coupling (e.g., nonlinear phase effects).

This equation is responsible for the emergence of directional structures and interference-like phenomena.

5.4 Evolution of the Kinetic Field K

The kinetic field evolves according to:

$$\begin{aligned} & K_{t+1} \\ &= \\ & K_t \\ &+ \kappa \nabla^2 K_t \\ &- \delta K_t \\ &+ P(J_t, W_t, H_t, R_t, \nabla X_t, \nabla^2 X_t). \end{aligned} \quad (13)$$

where P encodes feedback from dynamical consequences, including flux, vorticity, horizons, radiation, mantle gradients, and curvature-like deformation.

The production term typically takes the form:

$$P = \xi_J |J| + \xi_W |W| + \xi_H H + \xi_R R. \quad (14)$$

This expresses the central principle of OMM-SBT:

consequences feed back into the substrate.

5.5 Flux Coupling Term

The coupling between X and the flux structure is expressed through:

$$C_{\text{flux}} = \chi \nabla \cdot J, \quad (15)$$

or more generally:

$$C_{\text{flux}} = \chi_1 |J| + \chi_2 \nabla \cdot J. \quad (16)$$

This term allows propagation structures to influence the amplitude field.

5.6 Minimal Core Equation

A simplified version of the model, capturing its essential behavior, is given by:

$$X_{t+1} = X_t + D\nabla^2 X_t - \gamma X_t - \lambda X_t^3. \quad (17)$$

This minimal equation already produces:

- wave-like behavior,
- mode formation,
- and stable structures.

However, it does not include:

- phase dynamics,
- feedback,
- geometry,
- or thermodynamics.

5.7 Continuous Limit

In the continuous-time limit, the evolution equations take the form:

$$\partial_t X = D\nabla^2 X - \gamma X - \lambda X^3 + C_{\text{flux}} + C_{\text{finite}} + \eta, \quad (18)$$

$$\partial_t \Phi = \alpha \nabla^2 \Phi + \beta W + C_{\text{orient}}, \quad (19)$$

$$\partial_t K = \kappa \nabla^2 K - \delta K + P(J, W, H, R, \nabla X, \nabla^2 X). \quad (20)$$

5.8 Interpretative Summary

The dynamics of the OMM–SBT framework combine:

- diffusion (propagation),
- dissipation (relaxation),
- nonlinear saturation (stability),
- phase organization (coherence),
- and feedback (non-equilibrium dynamics).

This combination is sufficient to generate:

- wave-like sectors,
- flux-based interactions,

- curvature-like geometry,
- thermodynamic behavior,
- and boundary radiation effects.

The next section introduces a generating functional formulation, allowing these equations to be derived from an action-like principle.

6 Generating Functional and Action Principle

A central goal of any unifying physical framework is to express its dynamics in terms of a generating functional, typically an action, from which evolution equations can be derived.

In the OMM–SBT framework, this objective is approached in two complementary ways:

- a standard variational formulation, inspired by classical field theory,
- and an extended non-equilibrium functional incorporating feedback and kinetic effects.

6.1 Classical-Inspired Energy Functional

We begin with a classical energy-like functional for the amplitude field X :

$$E[X] = \int d^3x \left[\frac{D}{2} |\nabla X|^2 + \frac{\gamma}{2} X^2 + \frac{\lambda}{4} X^4 + \frac{\mu}{2} (\|X\|^2 - 1)^2 \right]. \quad (21)$$

Each term has a clear physical interpretation:

- $\frac{D}{2} |\nabla X|^2$: spatial gradient energy, governing diffusion and propagation;
- $\frac{\gamma}{2} X^2$: mass-like or damping term;
- $\frac{\lambda}{4} X^4$: nonlinear saturation, stabilizing the dynamics;
- $\frac{\mu}{2} (\|X\|^2 - 1)^2$: global constraint enforcing finite substrate behavior.

The conservative part of the X dynamics can be interpreted as a gradient-flow contribution:

$$\partial_t X = -\frac{\delta E}{\delta X} + \text{non-variational terms}. \quad (22)$$

The non-variational terms include flux coupling, normalization, stochastic excitation, and feedback. Therefore, this energy functional should not be interpreted as a complete action for the full model.

6.2 Inclusion of Phase and Flux Contributions

To incorporate directional and propagation effects, we extend the functional by including the phase field:

$$E[X, \Phi] = E[X] + \int d^3x \left[\frac{\alpha}{2} |\nabla \Phi|^2 + \chi |X| |\nabla \Phi| \right]. \quad (23)$$

Here:

- $\frac{\alpha}{2}|\nabla\Phi|^2$ promotes phase smoothness and coherence;
- $\chi|X||\nabla\Phi|$ couples amplitude and phase, generating the flux structure $J = |X|\nabla\Phi$.

This term introduces propagation pathways and is directly related to Maxwell-like behavior.

6.3 Extended Functional with Kinetic Feedback

A defining feature of OMM-SBT is the presence of kinetic feedback. This requires extending the functional beyond equilibrium descriptions.

We define a generalized functional:

$$\mathcal{A}[X, \Phi, K] = \int dt d^3x [\mathcal{L}_{\text{core}} + \mathcal{L}_{\text{flux}} + \mathcal{L}_{\text{kinetic}} + \mathcal{L}_{\text{feedback}}]. \quad (24)$$

The different contributions are:

6.3.1 Core Lagrangian

$$\mathcal{L}_{\text{core}} = \frac{D}{2}|\nabla X|^2 + \frac{\gamma}{2}X^2 + \frac{\lambda}{4}X^4. \quad (25)$$

6.3.2 Flux and Phase Contribution

$$\mathcal{L}_{\text{flux}} = \frac{\alpha}{2}|\nabla\Phi|^2 + \chi|X||\nabla\Phi|. \quad (26)$$

6.3.3 Kinetic Contribution

$$\mathcal{L}_{\text{kinetic}} = \frac{\kappa}{2}|\nabla K|^2 + \frac{\delta}{2}K^2. \quad (27)$$

This term controls the spatial distribution and relaxation of kinetic activity.

6.3.4 Feedback Contribution

$$\mathcal{L}_{\text{feedback}} = \eta_K K P(J, W, H, R, \nabla X, \nabla^2 X). \quad (28)$$

This term is non-standard and represents the key innovation of the framework. It encodes the fact that dynamical consequences (flux, vorticity, horizon, radiation, gradients, and curvature-like deformation) actively influence the substrate.

It explicitly breaks detailed balance and encodes the irreversible nature of feedback-driven dynamics.

6.4 Non-Equilibrium Character of the Action

Unlike standard field theories, the functional $\mathcal{A}$ is not purely variational in the classical sense. The presence of feedback implies that:

- the system is intrinsically non-equilibrium,
- the evolution cannot be fully reduced to a stationary action principle,
- additional dynamical rules (as given in Section 5) are required.

Thus, $\mathcal{A}$ should be interpreted as an effective generating functional for a driven dissipative system rather than as a fundamental least-action principle.

6.5 Relation to the Dynamical Equations

The equations of motion presented in Section 5 can be interpreted as:

- gradient flows of $\mathcal{A}$ for the conservative terms,
- augmented by explicit non-variational contributions for feedback and noise.

Symbolically:

$$\partial_t \Psi = -\frac{\delta \mathcal{A}}{\delta \Psi} + \text{non-variational terms.} \quad (29)$$

This structure is typical of driven dissipative systems and aligns with modern formulations of non-equilibrium field theory.

6.6 Minimal Generating Structure

A minimal version of the generating structure can be written as:

$$\mathcal{A}_{\min} = \int dt d^3x \left[\frac{D}{2} |\nabla X|^2 + \frac{\gamma}{2} X^2 + \frac{\lambda}{4} X^4 \right]. \quad (30)$$

This minimal form generates the core dynamics of X , but does not include:

- phase structure,
- feedback,
- thermodynamics,
- or geometry.

6.7 Interpretative Summary

The OMM–SBT action framework combines:

- classical energy-based structure,
- phase-induced propagation,
- kinetic accumulation,
- and feedback-driven non-equilibrium dynamics.

This hybrid formulation allows the model to:

- reproduce known physical behaviors,
- remain compatible with field-theoretic intuition,
- and extend beyond equilibrium-based theories.

The next sections show how structured physical phenomena emerge from this dynamical system.

7 Emergent Structures and Particle Interpretation

The OMM–SBT framework does not assume particles as primitive entities. Instead, all observable structures emerge dynamically from the coupled evolution of the amplitude field X , the phase field Φ , and the kinetic field K .

7.1 Wave Mantles and Distributed Structures

The primary structural element is the *wave mantle*, defined as a spatially extended region of coherent amplitude:

$$|X(x, t)| \gg \langle |X| \rangle, \quad (31)$$

with structured phase gradients.

Wave mantles:

- are diffuse and boundary-less,
- evolve continuously,
- carry both energy and orientation through (X, Φ) .

7.2 Diffuse Interaction Rings

When mantles overlap, they form diffuse interaction regions rather than sharp interfaces.

We define:

$$\mathcal{R}_{\text{int}} = \{x \mid |X| \text{ locally significant and } |\nabla X|, |J| \text{ locally enhanced}\}. \quad (32)$$

These regions:

- have finite and variable thickness,
- are dynamically modulated,
- can be asymmetric depending on global configuration.

They constitute **diffuse interaction rings**.

7.3 Proto-Particles as Energetic Stabilizations

A key refinement is that particles are not mantles themselves.

Instead:

$$\text{particle} \approx \text{meta-stable localized energetic stabilization within a diffuse interaction ring.} \quad (33)$$

These arise when:

- flux J converges,
- vorticity W is significant,
- kinetic feedback K stabilizes the configuration.

This explains:

- apparent localization,
- discrete energy levels,
- persistence over time.

7.4 Flux Skeleton and Propagation Channels

The flux field:

$$J = |X|\nabla\Phi, \quad (34)$$

defines a propagation skeleton.

It acts as:

- a transport network,
- a preferred direction for energy flow,
- the basis of Maxwell-like behavior.

7.5 Vorticity, Spirals, and Chirality

The vorticity field:

$$W = \nabla \times J, \quad (35)$$

generates rotational structures such as spirals and loops.

This leads to:

- chirality,
- rotational stability,
- Dirac/Pauli-like behavior.

7.6 Multi-Scale and Fractal Features

Nonlinear coupling and feedback produce multi-scale structures:

- hierarchical organization,
- fractal-like roughness,
- cascade processes.

7.7 Asymmetry and Context Dependence

Structures are generally asymmetric:

$$\nabla X(x) \neq -\nabla X(-x). \quad (36)$$

Their form depends on:

- flux configuration,
- neighboring mantles,
- kinetic feedback,
- boundary conditions.

7.8 Boundary Effects and Hidden Domains

Two types of boundaries exist:

- internal mantle boundaries (unobservable interiors),
- global substrate boundaries (finite-size effects).

Thus:

$$\text{observed physics} = \text{boundary-constrained projection of the substrate.} \quad (37)$$

7.9 Radiative Structures

Boundary regions generate radiation:

$$R = R(H, J, W, K, \nabla X, \nabla^2 X), \quad (38)$$

leading to:

- energy emission,
- redistribution of kinetic activity,
- Hawking-like behavior.

7.10 Interpretative Summary

In OMM-SBT:

- mantles are fundamental,
- interaction rings define structural interfaces,
- particles are stabilized energy patterns,
- matter is emergent,
- geometry and interactions arise from structure and flux.

8 Effective Geometry and Relativistic Behavior

In the OMM-SBT framework, geometry is not assumed as a fundamental structure. Instead, it emerges from the organization of the underlying fields (X, Φ, K) .

This section formalizes how effective geometry and relativistic behavior arise from the substrate dynamics.

8.1 Energy Density and Effective Geometry

We define an effective density:

$$\rho_{\text{eff}} = X^2 + \alpha K, \quad (39)$$

where:

- X is the mantle amplitude,

- K is the substrate kinetic field,
- α is a coupling coefficient.

This density captures both structural amplitude and kinetic agitation. The effective metric is then defined as:

$$g_{\text{eff}} = 1 + a\rho_{\text{eff}} + b|\nabla^2 X| + c|W| + d|J| + eK. \quad (40)$$

Each term contributes as follows:

- ρ_{eff} : energy density contribution,
- $|\nabla^2 X|$: curvature of the mantle,
- $|W|$: vorticity (rotational structure),
- $|J|$: flux intensity,
- K : kinetic feedback from the substrate.

8.2 Emergent Curvature

Spatial variations of g_{eff} define an effective curvature:

$$\mathcal{C}(x) = \nabla^2 g_{\text{eff}}(x). \quad (41)$$

Regions of high curvature correspond to:

- strong interactions,
- dense mantle overlap,
- high kinetic activity.

This provides an emergent analogue to gravitational curvature.

8.3 Relativistic Deformation as Structural Distortion

In this framework, relativistic effects correspond to deformations of the mantle structure and flux organization.

Specifically:

- time dilation emerges from slowed evolution of X and Φ in high-density regions,
- length contraction corresponds to anisotropic compression of flux pathways,
- mass corresponds to stabilized energy concentrations in interaction regions.

Thus:

$$\text{relativity} \approx \text{effective structural response of the substrate to energy density and flux organization.} \quad (42)$$

8.4 Causal Structure and Horizons

The finite nature of the substrate introduces causal boundaries defined by:

$$H = \{x \mid g_{\text{eff}}(x) > q_{\theta}(g_{\text{eff}})\}. \quad (43)$$

These horizons:

- separate accessible and inaccessible regions,
- define causal limits of propagation,
- correspond to effective horizons similar to black hole boundaries.

8.5 Radiation from Boundary Dynamics

At these boundaries, the system generates radiation through imbalance between interior and exterior dynamics:

$$R = R(H, J, W, K, \nabla X, \nabla^2 X). \quad (44)$$

This leads to:

- Hawking-like radiation,
- energy leakage from confined regions,
- redistribution of kinetic energy.

8.6 Particle Interpretation Revisited

In light of the structural framework, particles are not primitive objects.

Instead:

$$\text{particle} \approx \text{meta-stable localized energetic stabilization within diffuse interaction rings}. \quad (45)$$

These occur at:

- intersections of wave mantles,
- regions of high flux convergence,
- zones of enhanced vorticity and kinetic feedback.

This explains:

- apparent localization,
- discrete energy levels,
- interaction-dependent properties.

8.7 Observability and Hidden Structure

Due to boundary effects:

- inner mantle structure is not directly observable,
- global substrate structure beyond the observable domain is inaccessible,
- physical measurements correspond to boundary projections.

Thus:

$$\text{observed geometry} = \text{effective projection of a deeper dynamic substrate.} \quad (46)$$

8.8 Interpretative Summary

In OMM–SBT:

- geometry is emergent,
- curvature is induced by energy organization,
- relativity is a dynamical structural effect,
- horizons arise from finite-size constraints,
- particles are interaction-localized energy phenomena.

This establishes a unified interpretation of geometry, matter, and interaction within a single dynamical framework.

9 Horizons and Boundary Effects

In OMM–SBT, horizons are not introduced as fundamental spacetime surfaces. They emerge from finite-size constraints and from extreme values of the effective geometry.

We define a horizon-like region as:

$$H = \{x \mid g_{\text{eff}}(x) > q_{\theta}(g_{\text{eff}})\}.$$

Such regions behave as causal boundaries because propagation, feedback, and radiation are modified near them.

Boundary effects appear at two levels:

- local mantle boundaries, associated with interaction rings;
- global substrate boundaries, associated with the finite observable domain.

Thus, observed physics corresponds to a boundary-constrained projection of the deeper substrate:

$$\text{observed physics} = \text{projection of substrate dynamics through accessible boundaries.}$$

Radiation-like behavior appears when kinetic activity, flux, vorticity, curvature, and boundary structure become imbalanced:

$$R = R(H, J, W, K, \nabla X, \nabla^2 X).$$

This provides the basis for Hawking-like behavior in the model.

10 Generalized Continuity

Because OMM–SBT includes global normalization, finite-size constraints, and consequence-to-substrate feedback, local conservation is not expected to take the classical form

$$\Delta_t \rho + \nabla \cdot J = 0.$$

Instead, the model predicts a generalized continuity relation:

$$\Delta_t \rho_{\text{eff}} + \nabla \cdot J_{\text{eff}} = S_{\text{total}}.$$

Here,

$$\rho_{\text{eff}} = X^2 + \alpha_K K$$

is the effective density, while

$$J_{\text{eff}} = |X| \nabla \Phi + \beta_K \nabla K$$

is the effective current.

The source term is decomposed as:

$$S_{\text{total}} = S_{\text{norm}} + S_{\text{feedback}} + S_{\text{boundary}} + S_K + S_W + S_J + S_R.$$

This term does not represent arbitrary violation of conservation, but arises systematically from normalization constraints, feedback processes, and boundary effects.

Thus, conservation is generalized:

$$\text{local non-conservation} \quad \leftrightarrow \quad \text{global constrained conservation}.$$

This is one of the central structural predictions of OMM–SBT.

11 Thermodynamics and Arrow of Time

Thermodynamic behavior in the OMM–SBT framework does not arise from a single isolated sector. Rather, it emerges as a global kinetic projection of the full substrate dynamics.

The kinetic field K integrates contributions from flux, vorticity, boundary activity, radiation, curvature-like deformation, and feedback. In this sense, thermodynamics is not an additional layer added to the model, but the collective expression of the coupled sectors.

11.1 Substrate Kinetics

We introduce a kinetic field $K(x, t)$ representing local agitation of the substrate.

This field:

- stores the cumulative effect of interactions,
- evolves through diffusion and dissipation,
- feeds back into the dynamics of X and Φ .

Its evolution is given by:

$$K_{t+1} = K_t + \kappa \nabla^2 K_t - \delta K_t + P(J, W, H, R, \nabla X, \nabla^2 X), \quad (47)$$

where P encodes the cumulative kinetic contribution of flux, vorticity, horizons, radiation, and mantle gradients.

11.2 Effective Temperature

We define an effective temperature as:

$$T_{\text{eff}} \propto K. \quad (48)$$

This reflects:

- local agitation,
- interaction intensity,
- energy redistribution.

11.3 Entropy as Distribution of Kinetic Activity

Entropy is interpreted as the spread of kinetic activity across the substrate.

We define:

$$S \sim \int K(x, t) \log K(x, t) dx. \quad (49)$$

This captures:

- dispersion of energy,
- loss of localization,
- increase of configurational complexity.

11.4 Non-Equilibrium Dynamics

The system is fundamentally out of equilibrium due to:

- continuous excitation (battement substantiel),
- feedback from emergent structures,
- boundary effects (finite substrate).

Thus:

$$\text{equilibrium is never fully reached.} \quad (50)$$

11.5 Arrow of Time

The arrow of time emerges from:

- irreversible redistribution of K ,
- accumulation of feedback,
- asymmetry in boundary interactions.

Formally:

$$\frac{dS}{dt} \geq 0. \quad (51)$$

This is not imposed but emerges from:

- dissipation,
- diffusion,
- feedback loops.

11.6 Thermodynamic Origin of Radiation

Radiation arises from imbalances at causal boundaries:

$$R = R(H, J, W, K, \nabla X, \nabla^2 X). \quad (52)$$

This leads to:

- energy emission,
- entropy increase,
- coupling between interior and exterior regions.

This provides a thermodynamic basis for Hawking-like radiation.

11.7 Feedback and Thermodynamic Closure

A key feature of the model is feedback:

$$\text{consequences} \rightarrow K \rightarrow \text{new dynamics}. \quad (53)$$

This creates:

- self-organized structures,
- persistent non-equilibrium states,
- thermodynamic loops.

For this reason, the thermodynamic sector should be interpreted as robustly emergent rather than fully analytically closed. While numerical validation shows derived-candidate behavior, a complete analytic derivation requires solving the fully coupled system of fields (X, Φ, K) .

11.8 Interpretative Summary

In OMM–SBT:

- temperature is kinetic agitation,
- entropy is distribution of kinetic activity,
- time asymmetry emerges from feedback and dissipation,
- radiation is a boundary thermodynamic effect,
- thermodynamics is a global kinetic projection of all coupled sectors.

12 Derivation of Physical Sectors

One of the central claims of the OMM–SBT framework is that distinct physical theories arise as limiting regimes of a single underlying causal-kinetic substrate.

In this section, we show how the main physical sectors emerge from the fundamental fields (X, Φ, K) and their coupled dynamics.

12.1 General Strategy

The derivation of sectors follows a common principle:

- identify a dominant regime (diffusion, phase, curvature, boundary, etc.),
- simplify the full dynamics accordingly,
- extract the effective equation governing observable behavior.

Thus:

$$\text{sector} \approx \text{limiting regime of the full dynamics.} \quad (54)$$

12.2 Schrödinger-like Sector

In regimes dominated by diffusion and weak nonlinearity, the amplitude field evolves as:

$$X_{t+1} \approx X_t + D\nabla^2 X_t - \gamma X_t. \quad (55)$$

Passing to a continuous limit:

$$\partial_t X = D\nabla^2 X - \gamma X. \quad (56)$$

Coupled with the phase field Φ , this leads to wave-like behavior analogous to the Schrödinger equation.

Interpretation:

- X plays the role of amplitude,
- Φ encodes phase evolution,
- interference and dispersion arise naturally.

12.3 Dirac / Pauli-like Sector

The Dirac-like behavior emerges from orientation and vorticity.

The flux:

$$J = |X|\nabla\Phi, \quad (57)$$

combined with vorticity:

$$W = \nabla \times J, \quad (58)$$

produces chiral, two-component structures.

Interpretation:

- spinor-like behavior arises from orientation dynamics,
- chirality is linked to rotational structure,
- two-component coupling is encoded in phase-vorticity interaction.

12.4 Maxwell-like Sector

The flux field acts as an effective field:

$$J = |X|\nabla\Phi. \quad (59)$$

In regimes where amplitude varies slowly, we obtain:

$$\nabla \cdot J \approx \text{source terms}, \quad \nabla \times J = W. \quad (60)$$

This yields a Maxwell-like structure:

- divergence $\rightarrow$ charge-like effects,
- curl $\rightarrow$ field circulation,
- propagation along flux lines.

12.5 Einstein-like Sector

Geometry emerges through the effective metric:

$$g_{\text{eff}} = 1 + a\rho + b|\nabla^2 X| + c|W| + d|J| + eK. \quad (61)$$

Energy density and structure modify geometry:

- high $|X| \rightarrow$ effective curvature,
- strong vorticity $\rightarrow$ rotational deformation,
- kinetic activity $K \rightarrow$ dynamic geometry.

Thus:

$$\text{geometry} \approx \text{functional of energy distribution}. \quad (62)$$

12.6 Thermodynamic Sector

The kinetic field K encodes substrate agitation as a global projection of the coupled sectors:

$$K_{t+1} = K_t + \kappa\nabla^2 K_t - \delta K_t + P(J, W, H, R, \nabla X, \nabla^2 X). \quad (63)$$

We identify:

$$T_{\text{eff}} \propto K. \quad (64)$$

Thus:

- temperature arises from kinetic activity,
- entropy corresponds to the distribution of kinetic activity,
- dissipation emerges from relaxation and feedback terms.

Thermodynamics is therefore robustly emergent and numerically reaches derived-candidate behavior, but its full analytic closure remains a global problem involving all coupled sectors.

12.7 Hawking-like and Radiation Sector

At boundaries defined by:

$$H = \{g_{\text{eff}} > q\theta(g_{\text{eff}})\}, \quad (65)$$

flux asymmetry and kinetic imbalance produce radiation:

$$R = R(H, J, W, K, \nabla X, \nabla^2 X). \quad (66)$$

Interpretation:

- boundary-induced energy emission,
- kinetic imbalance across horizons,
- analogue of Hawking-like radiation.

12.8 String/M-like Sector

Extended structures arise as low-frequency modes of the mantle field.

These correspond to:

- coherent extended oscillations,
- stable filament-like structures,
- effective higher-order degrees of freedom.

Thus:

$$\text{extended structures} \approx \text{collective modes of } X. \quad (67)$$

12.9 Unified Interpretation

All sectors emerge from the same underlying dynamics:

$$\Psi_t = (X_t, \Phi_t, K_t). \quad (68)$$

Each physical theory corresponds to:

- a regime of dominance,
- a projection of the full dynamics,
- a simplification of coupled interactions.

12.10 Summary

OMM-SBT provides a unified origin for:

- quantum behavior (diffusion + phase),
- spinor dynamics (vorticity + orientation),
- electromagnetic fields (flux structure),
- gravity (emergent geometry),

- thermodynamics (kinetic substrate),
- radiation (boundary effects),
- extended modes (string-like structures).

This supports the interpretation of the framework as a coherent unification of physical sectors.

13 Validation and Numerical Evidence

The OMM-SBT framework has been subjected to a progressive numerical validation program, designed to test both internal coherence and correspondence with known physical behaviors.

This section summarizes the validation strategy and key results obtained across multiple iterations of the model.

13.1 Validation Strategy

The validation process follows three complementary axes:

- **Calibration:** progressive refinement of model parameters and structures (V1 $\rightarrow$ V116),
- **Independent reproduction:** verification using a separate implementation (V115),
- **Invariance testing:** assessment of fundamental symmetries (V113).

Together, these form a minimal but coherent validation framework.

13.2 Progressive Calibration (V1 $\rightarrow$ V116)

The model was iteratively refined through successive versions:

- early stages: identification of stable dynamics (flux, vorticity, phase),
- intermediate stages: emergence of sector-like behaviors,
- advanced stages: introduction of feedback, horizons, and generalized continuity.

Key outcomes include:

- stabilization of flux structures,
- emergence of vorticity-driven chirality,
- formation of effective horizons,
- appearance of radiation-like boundary behavior,
- convergence toward a unified causal-kinetic structure.

This progressive refinement ensures that the final model is not arbitrarily constructed, but emerges from systematic exploration.

13.3 Independent Reproduction

A crucial validation step consists in reproducing the model using a separate, simplified implementation.

This independent version:

- does not reuse the original exploratory pipeline,
- relies only on the core equations,
- tests robustness of emergent behavior.

The public reference implementation shows:

- 9/9 axes exhibit robust behavior,
- 8/9 axes reach derived-candidate level,
- a consistent core score ~ 0.88 across configurations.

These include:

- Schrödinger-like behavior (diffusion + phase),
- Dirac/Pauli-like structures (vorticity),
- Maxwell-like flux organization,
- Einstein-like effective geometry,
- thermodynamic behavior as a global kinetic projection,
- Hawking-like boundary radiation,
- radiation coherence,
- robust generalized continuity improvement.

This strongly supports the structural robustness of the model.

13.4 Invariance Scan (V113)

Fundamental symmetries were tested through dedicated invariance scans.

The following symmetries show strong validation:

- translation invariance,
- rotational invariance,
- reflection symmetry,
- global phase invariance,
- local gauge stability,
- Lorentz-boost proxy behavior,

- Bell-type bounded correlations,
- Pauli/spinor-like structure.

Quantitatively:

$$\text{mean invariance score} \approx 0.94 - 1.00 \quad (\text{strong regime}). \quad (69)$$

One limitation remains:

- strict local continuity remains partially violated in naive form,
- which motivated the introduction of a generalized continuity equation.

13.5 Generalized Continuity Validation

The standard conservation equation:

$$\partial_t \rho + \nabla \cdot J = 0 \quad (70)$$

does not hold strictly due to:

- global substrate constraints,
- boundary effects,
- feedback mechanisms.

A generalized form was introduced:

$$\Delta_t \rho_{\text{eff}} + \nabla \cdot J_{\text{eff}} = S_{\text{total}}. \quad (71)$$

Validation results indicate:

- consistent reduction of local residual errors,
- improvement ratios up to ~ 0.6 ,
- strong correlation between modeled and observed residuals.

This supports the interpretation of S_{total} as a physically meaningful source term.

13.6 Thermodynamic and Radiation Validation

The kinetic field K successfully reproduces:

- temperature-like behavior,
- entropy distribution patterns,
- relaxation dynamics.

These results are robust across parameter variations and reach derived-candidate level in numerical experiments.

However, thermodynamics should be interpreted as a global emergent projection of the coupled sectors rather than a fully independent analytically derived sector.

Boundary regions produce:

- asymmetric flux,
- radiation-like emission,
- Hawking-like signatures.

These behaviors are consistently observed across configurations.

13.7 Limitations of Current Validation

Despite strong results, several limitations must be acknowledged:

- continuity is not strictly conserved in local form,
- analytic closure remains incomplete,
- Lorentz invariance is validated only through proxy tests,
- thermodynamics is numerically robust but not yet analytically closed as an independent sector.

These limitations do not invalidate the framework, but define clear directions for future work.

13.8 Summary of Validation Results

The validation program supports the following conclusions:

- the model is internally coherent,
- emergent structures are robust,
- multiple physical sectors arise consistently,
- fundamental symmetries are largely respected,
- deviations (e.g., continuity) are explainable within the framework.

Overall, the results strongly support the interpretation of OMM–SBT as a viable unified physical framework candidate.

14 Falsifiable Predictions

A defining requirement of a physical theory is its ability to produce clear, testable, and potentially falsifiable predictions.

The OMM–SBT framework leads to several such predictions, directly rooted in its core structures: flux J , vorticity W , kinetic field K , and boundary dynamics.

14.1 Prediction 1: Flux-Driven Anisotropic Propagation

In OMM–SBT, propagation is structured by the flux field:

$$J = |X|\nabla\Phi. \quad (72)$$

Unlike isotropic wave propagation, this defines preferred directions.

Prediction:

- measurable anisotropy in propagation speed or intensity,
- alignment of propagation patterns with flux structures,
- deviation from isotropic dispersion relations.

Falsification criterion: Strict isotropic propagation in systems where flux structures are expected would contradict the model.

14.2 Prediction 2: Finite-Width Interaction Regions

Interactions arise from mantle overlap and flux convergence, forming diffuse regions rather than point-like events.

Prediction:

- interaction zones with finite, measurable thickness,
- thickness scaling with field gradients or system energy,
- spatially extended collision regions in wave-like systems.

Falsification criterion: Observation of strictly point-like interactions at all accessible scales would challenge the framework.

14.3 Prediction 3: Kinetic Contribution to Temperature

Temperature is linked to substrate kinetics:

$$T_{\text{eff}} \propto K. \quad (73)$$

Prediction:

- correlation between dynamic field activity and effective temperature,
- deviations from equilibrium thermodynamics in strongly driven systems,
- temperature anomalies linked to structural reconfiguration.

Falsification criterion: Absence of any correlation between dynamic field activity and thermal behavior in non-equilibrium regimes.

14.4 Prediction 4: Boundary-Induced Radiation

Finite boundaries (horizons) generate radiation through kinetic imbalance and feedback:

$$R = R(H, J, W, K, \nabla X, \nabla^2 X). \quad (74)$$

Prediction:

- energy emission localized at high-curvature or high-gradient boundaries,
- dependence of radiation intensity on boundary geometry,
- Hawking-like spectra in effective horizon systems.

Falsification criterion: Absence of any boundary-dependent emission in systems with strong confinement or horizon-like structures.

14.5 Prediction 5: Context-Dependent Particle Stability

Particles correspond to stabilized energetic configurations within interaction regions.

Prediction:

- stability thresholds depending on flux and vorticity,
- sensitivity of particle-like states to environmental perturbations,
- disappearance or transformation under structural disruption.

Falsification criterion: Absolute intrinsic stability of particles independent of interaction context.

14.6 Prediction 6: Vorticity-Induced Chirality and Spin Structures

The vorticity field:

$$W = \nabla \times J \quad (75)$$

induces rotational and chiral organization.

Prediction:

- emergence of spin-like behavior from rotational flux structures,
- chirality linked to vorticity orientation,
- correlation between rotational patterns and two-component (spinor-like) dynamics.

Falsification criterion: Complete absence of rotational or chiral structure in systems governed by flux dynamics.

14.7 Prediction 7: Scale-Coupled (Fractal-Like) Structure Formation

Nonlinear coupling and feedback generate multi-scale organization.

Prediction:

- hierarchical structures across scales,
- scale-dependent roughness or fractal signatures,
- cascade-like redistribution of energy.

Falsification criterion: Strict scale independence in systems where nonlinear coupling is dominant.

14.8 Experimental Outlook

These predictions can be investigated in:

- wave-based analog systems (optical, acoustic, fluid),
- nonlinear field simulations,
- condensed matter systems with strong collective behavior,
- controlled boundary systems (cavities, traps, metamaterials).

The combination of multiple validated predictions would provide strong support for the OMM–SBT framework, while clear violations would allow decisive falsification.

15 Comparison with Existing Theoretical Frameworks

The OMM–SBT framework proposes a unified description of physical phenomena based on a dynamical substrate composed of amplitude, phase, and kinetic fields.

This section compares the framework with major existing theories, highlighting both correspondences and differences.

15.1 Comparison with Quantum Mechanics

In standard quantum mechanics:

- the wavefunction ψ encodes probability amplitudes,
- particles exhibit wave-particle duality,
- observables arise through measurement.

In OMM–SBT:

- the amplitude field X plays a role analogous to $|\psi|$,
- phase Φ provides directional and dynamical information,
- particle-like behavior emerges from localized energy stabilizations.

Key differences:

- no fundamental probabilistic postulate is required,
- no collapse mechanism is assumed,
- localization arises from dynamics rather than measurement.

15.2 Comparison with Quantum Field Theory

Quantum Field Theory (QFT) describes particles as excitations of underlying fields.

OMM-SBT shares this perspective:

- structures are excitations of a continuous substrate,
- interactions emerge from field coupling.

However:

- the substrate is explicitly dynamical and finite,
- feedback through the kinetic field K is intrinsic,
- interaction regions are diffuse rather than point-like.

15.3 Comparison with General Relativity

General Relativity (GR) describes gravity as curvature of spacetime.

In OMM-SBT:

- curvature emerges from the effective metric:

$$g_{\text{eff}} = 1 + a\rho + b|\nabla^2 X| + c|W| + d|J| + eK, \quad (76)$$

- energy distribution determines geometric deformation,
- horizons arise from finite-size constraints.

Key distinction:

- geometry is not fundamental but emergent,
- spacetime is replaced by a dynamical substrate,
- gravitational effects arise from structural organization.

15.4 Comparison with Thermodynamics

Classical thermodynamics treats temperature and entropy as macroscopic quantities.

In OMM-SBT:

- temperature is linked to substrate kinetics:

$$T_{\text{eff}} \propto K, \quad (77)$$

- entropy corresponds to the distribution of kinetic activity,
- irreversibility emerges from feedback and dissipation.

Thus:

- thermodynamics is not introduced as an external layer,
- it emerges as a global kinetic projection of the coupled substrate dynamics.

This thermodynamic sector is numerically robust and reaches derived-candidate behavior. Its incomplete analytic closure reflects the fact that thermodynamics depends on the full set of coupled sectors, rather than constituting an isolated subsystem.

15.5 Comparison with String Theory and M-Theory

String theory models fundamental entities as one-dimensional objects in higher-dimensional spaces.

OMM-SBT differs fundamentally:

- no additional spatial dimensions are required,
- extended structures arise as effective modes of the substrate,
- string-like behavior emerges from low-frequency mantle structures.

Thus:

$$\text{extended structures} \approx \text{kinetic modes, not extra dimensions.} \quad (78)$$

15.6 Common Ground Across Theories

Despite differences, several common themes appear:

- fields as fundamental entities (QFT, OMM-SBT),
- emergence of structure from underlying dynamics,
- geometry linked to energy (GR, OMM-SBT),
- extended modes (string theory, OMM-SBT).

15.7 Key Distinctive Features of OMM-SBT

The framework introduces several original elements:

- explicit finite substrate,
- intrinsic feedback loop (consequence $\rightarrow$ substrate),
- unified description of geometry, matter, and thermodynamics,
- absence of primitive particles,
- 4D sufficiency without additional dimensions.

15.8 Interpretative Position

OMM-SBT can be interpreted as:

- a causal-kinetic field theory,
- an emergent unification framework,
- a candidate bridge between quantum, relativistic, and thermodynamic descriptions.

It does not claim equivalence with existing theories, but rather proposes a unified underlying structure from which they may arise as limits.

15.9 Summary

OMM-SBT aligns with established theories at the level of observable phenomena, while differing in its ontological assumptions and dynamical mechanisms.

It replaces:

- particles by interaction-localized energy patterns,
- spacetime by an emergent effective geometry,
- thermodynamics by intrinsic substrate kinetics.

This positions the framework as a serious candidate for a unified description of physical reality.

16 Discussion

The OMM-SBT framework proposes a unified, causal-kinetic description of physical phenomena, in which all observed structures emerge from a common oscillatory substrate.

This section situates the framework with respect to existing theories, clarifies its contributions, and outlines its current limitations.

16.1 Relation to Quantum Mechanics

The emergence of Schrödinger-like behavior from diffusion and phase dynamics:

$$J = |X|\nabla\Phi \tag{79}$$

suggests that wave mechanics is not fundamental, but arises as an effective description of underlying field dynamics.

In this perspective:

- the wavefunction corresponds to a reduced description of (X, Φ) ,
- probabilistic interpretation may reflect unresolved kinetic contributions K ,
- non-local correlations are linked to extended mantle structures.

This provides a potential reinterpretation of quantum mechanics as an emergent statistical framework.

16.2 Relation to Relativity

Effective geometry arises from structural properties of the substrate:

$$g_{\text{eff}} = 1 + a\rho + b|\nabla^2 X| + c|W| + d|J| + eK. \tag{80}$$

This suggests that:

- curvature is not fundamental but emergent,
- spacetime geometry reflects field organization,
- gravitational effects arise from energy distribution and structure.

While this reproduces Einstein-like behavior, it does not yet constitute a full derivation of General Relativity.

16.3 Relation to Thermodynamics

The introduction of the kinetic field K allows a unified treatment of thermodynamics:

$$T_{\text{eff}} \propto K. \quad (81)$$

This leads to:

- a natural description of non-equilibrium systems,
- entropy as a structural property of energy distribution,
- thermodynamic behavior emerging from feedback dynamics.

Thermodynamics should therefore be interpreted as a global kinetic projection of the coupled substrate dynamics rather than an isolated sector. Its numerical robustness is established, while its incomplete analytic closure reflects its dependence on all interacting components of the model.

16.4 Relation to Field Theory and Gauge Structures

The strong invariance results suggest compatibility with field-theoretic structures:

- global phase invariance,
- local gauge stability,
- conservation-like structures (generalized continuity).

The presence of a non-zero source term:

$$\Delta_t \rho_{\text{eff}} + \nabla \cdot J_{\text{eff}} = S_{\text{total}} \quad (82)$$

indicates that conservation laws may be emergent and constrained rather than fundamental.

This departs from standard formulations and requires further formal clarification.

16.5 Relation to String and M-Theory

The emergence of extended structures (mantles, flux tubes, rings) provides a natural analogy with string-like objects.

However:

- no extra spatial dimensions are introduced,
- extended objects arise dynamically rather than being postulated,
- no direct equivalence with Polyakov or Nambu–Goto actions is claimed.

In this sense, OMM–SBT may be interpreted as a lower-dimensional, kinetic realization of extended structures.

16.6 Conceptual Contributions

The framework introduces several conceptual shifts:

- particles are not fundamental entities,
- matter corresponds to stabilized energy configurations,
- interactions are diffuse and structured,
- causality includes feedback from consequences to substrate.

This leads to a fully relational and dynamic ontology.

16.7 Limitations and Open Problems

Despite promising results, several limitations remain:

- the generalized continuity equation is not yet fully derived,
- analytic closure of the system is incomplete,
- Lorentz invariance is supported by proxy tests rather than formal proof,
- thermodynamic sector remains globally coupled and not fully isolated analytically,
- quantization is not yet rigorously established.

These issues define a clear research program for future work.

16.8 Position as a ToE Candidate

The combination of:

- unified field description,
- emergence of multiple physical sectors,
- strong invariance properties,
- independent numerical reproduction,

supports the interpretation of OMM–SBT as a serious candidate for an emergent unification framework.

However, the use of the term "Theory of Everything" should remain conditional on:

- external validation,
- analytic completion,
- experimental confirmation.

16.9 Summary

OMM–SBT provides a coherent and extensible framework that:

- unifies multiple physical phenomena under a single substrate,
- reproduces key structural aspects of known theories,
- introduces new perspectives on causality, matter, and interaction,
- remains open to refinement and verification.

17 Conclusion

This work introduces the Oscillatory Mantle Model – Substantial Beating Theory (OMM–SBT) as a unified causal-kinetic framework for physical phenomena.

The central hypothesis is that all observable structures emerge from a common oscillatory substrate, described by the coupled evolution of amplitude X , phase Φ , and kinetic field K . Within this framework, particles, interactions, geometry, and thermodynamics are not fundamental entities, but consequences of structured field dynamics.

The model successfully reproduces multiple physical sectors as emergent limits:

- wave dynamics consistent with Schrödinger-like behavior,
- flux-based propagation analogous to Maxwell theory,
- vorticity-driven structures related to Dirac/Pauli features,
- effective geometry resembling Einstein-like curvature,
- thermodynamic behavior linked to substrate kinetics,
- boundary-induced radiation analogous to Hawking processes.

These results are supported by:

- progressive calibration across multiple model versions,
- strong invariance properties,
- and independent numerical reproduction of core behaviors.

A key conceptual contribution of OMM–SBT is the reinterpretation of matter as a stabilized energetic phenomenon arising within diffuse interaction regions, rather than as a fundamental entity. In this view, causality is not strictly unidirectional, but includes feedback from emergent consequences to the underlying substrate.

Despite these advances, several aspects remain incomplete:

- analytic closure of the generalized continuity equation,
- full analytic closure of thermodynamics as a globally coupled sector,
- formal demonstration of Lorentz invariance,

- and a complete quantization procedure.

These limitations define a clear path for future work.

From a broader perspective, the OMM–SBT framework suggests that physical laws may not be fundamental constraints imposed on reality, but emergent regularities arising from the organization of a deeper dynamical substrate.

In this sense, the present work should be understood not as a final Theory of Everything, but as a structured and testable candidate for emergent unification.

Its validity ultimately depends on:

- external reproduction,
- confrontation with experimental observations,
- and further analytical development.

If confirmed, this framework would provide a new perspective on the nature of matter, interaction, and spacetime, grounded in a unified causal-kinetic description.

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A Complete Set of Model Equations

This appendix summarizes the full mathematical structure of the OMM–SBT framework.

A.1 State Variables

The complete state of the system is:

$$\Psi_t = (X_t, \Phi_t, K_t),$$

where:

- X_t is the oscillatory mantle amplitude,
- Φ_t is the phase/orientation field,
- K_t is the substrate kinetic field.

A.2 Derived Quantities

The effective density is:

$$\rho_{\text{eff}} = X^2 + \alpha_K K.$$

The flux skeleton is:

$$J = |X| \nabla \Phi.$$

The vorticity is:

$$W = \nabla \times J.$$

The curvature proxy is:

$$C = |\nabla^2 X|.$$

The effective metric is:

$$g_{\text{eff}} = 1 + a\rho_{\text{eff}} + bC + c|W| + d|J| + eK.$$

The horizon-like region is:

$$H = \{x \mid g_{\text{eff}}(x) > q_\theta(g_{\text{eff}})\}.$$

The radiation proxy is:

$$R = R(H, J, W, K, \nabla X, \nabla^2 X).$$

A.3 Full Discrete Evolution

The mantle amplitude evolves as:

$$X_{t+1} = \mathcal{N} [X_t + D\nabla^2 X_t - \gamma X_t - \lambda X_t^3 + C_{\text{flux}} + C_{\text{finite}} + \eta_t].$$

The finite-substrate correction is:

$$C_{\text{finite}} = -\mu(\|X_t\|^2 - 1)X_t.$$

The flux correction may be written:

$$C_{\text{flux}} = \chi_1 |J_t| + \chi_2 \nabla \cdot J_t.$$

The phase evolves as:

$$\Phi_{t+1} = \Phi_t + \alpha \nabla^2 \Phi_t + \beta W_t + C_{\text{orient}}.$$

The substrate kinetic field evolves as:

$$K_{t+1} = K_t + \kappa \nabla^2 K_t - \delta K_t + P(J_t, W_t, H_t, R_t, \nabla X_t, \nabla^2 X_t).$$

A typical production term is:

$$P(J, W, H, R, \nabla X, \nabla^2 X) = \xi_J |J| + \xi_W |W| + \xi_H H + \xi_R R.$$

A.4 Minimal Equation

The minimal mantle equation is:

$$X_{t+1} = X_t + D\nabla^2 X_t - \gamma X_t - \lambda X_t^3.$$

This minimal form captures diffusion, dissipation, nonlinear saturation, and attractor formation, but does not include phase, feedback, geometry, or thermodynamics.

A.5 Continuous Limit

In the continuous-time limit:

$$\partial_t X = D\nabla^2 X - \gamma X - \lambda X^3 + C_{\text{flux}} + C_{\text{finite}} + \eta.$$

$$\partial_t \Phi = \alpha \nabla^2 \Phi + \beta W + C_{\text{orient}}.$$

$$\partial_t K = \kappa \nabla^2 K - \delta K + P(J, W, H, R, \nabla X, \nabla^2 X).$$

A.6 Generalized Continuity

The generalized continuity relation is:

$$\Delta_t \rho_{\text{eff}} + \nabla \cdot J_{\text{eff}} = S_{\text{total}}.$$

The effective current is:

$$J_{\text{eff}} = |X| \nabla \Phi + \beta_K \nabla K.$$

The source term is:

$$S_{\text{total}} = S_{\text{norm}} + S_{\text{feedback}} + S_{\text{boundary}} + S_K + S_W + S_J + S_R.$$

This expresses local non-conservation induced by global substrate constraints and feedback dynamics.

A.7 Generating Functional

A candidate generating functional is:

$$\mathcal{A} = \int dt d^3x \left[\frac{D}{2} |\nabla X|^2 + \frac{\gamma}{2} X^2 + \frac{\lambda}{4} X^4 + \frac{\mu}{2} (\|X\|^2 - 1)^2 + \Omega(J, W, H, K, \nabla X, \nabla^2 X) \right].$$

The additional term Ω collects flux, vorticity, boundary, and kinetic feedback contributions.

A.8 Diffuse Interaction Rings

Diffuse Interaction Rings are defined as interaction regions satisfying:

$$\mathcal{R}_{\text{int}} = \{x \mid \text{mantle overlap and flux convergence}\}.$$

Particle-like structures correspond to:

$$\text{particle} \approx \text{localized energetic stabilization within } \mathcal{R}_{\text{int}}.$$

B Reproducibility and Implementation

This appendix provides the necessary elements to reproduce the results presented in this work.

B.1 Code Availability

A complete implementation of the OMM-SBT framework is provided in a public repository:

- Repository: <https://github.com/NicolasGilbertAlbertRoux/omm-sbt-causal-kinetic-framework.git>
- Language: Python
- Structure: modular scripts for simulation, validation, and analysis

The repository includes:

- core evolution equations for (X, Φ, K) ,
- validation pipelines (V1 to V116),
- independent reproduction implementation (V115),
- data generation and analysis scripts.

B.2 Execution Protocol

A typical execution follows these steps:

1. initialize the fields X , Φ , and K ,
2. run the evolution equations over a fixed number of iterations,
3. compute derived quantities:
 - flux J ,

- vorticity W ,
 - effective metric g_{eff} ,
 - horizon regions H ,
 - radiation indicators R ,
4. evaluate sector-specific scores,
 5. aggregate results into validation summaries.

B.3 Parameter Configuration

Typical parameter ranges used in the experiments are:

- diffusion coefficient $D \in [0.10, 0.20]$,
- damping coefficient $\gamma \in [0.006, 0.012]$,
- nonlinear coefficient $\lambda \in [0.02, 0.035]$,
- kinetic coupling $\kappa \in [0.08, 0.16]$,
- feedback strength $\chi \in [0.02, 0.06]$.

These values are not fine-tuned to a single configuration, but chosen within stable regions of the model.

B.4 Independent Reproduction

A simplified implementation is provided to verify that the core results do not depend on the original codebase.

This independent version:

- uses minimal equations,
- avoids reuse of the main pipeline,
- reproduces the main sector behaviors.

This step is essential to ensure robustness and reduce implementation bias.

B.5 Output and Validation Metrics

The following metrics are computed:

- sector scores (Schrödinger, Dirac, Maxwell, Einstein, etc.),
- invariance measures,
- continuity residuals,
- radiation and boundary indicators,
- global coherence metrics.

Results are exported as structured data (CSV files) for analysis.

B.6 Reproducibility Scope

All numerical results presented in this work can be reproduced using:

- the provided codebase,
- the documented parameter ranges,
- standard computational resources.

No proprietary tools or inaccessible datasets are required.

B.7 Limitations

Reproducibility is currently limited by:

- discretization choices,
- finite simulation domains,
- absence of fully continuous analytic formulation.

These limitations do not affect the qualitative conclusions but may impact precise quantitative values.

B.8 Future Improvements

Future work will include:

- optimized implementations,
- higher-resolution simulations,
- standardized benchmarks,
- cross-platform validation.

C Numerical Protocol

This appendix details the numerical methodology used to generate and validate the results of the OMM-SBT framework.

C.1 Discretization

The system is simulated on a discrete spatial grid:

- lattice size: typically $N = 48$ to 80 ,
- boundary conditions: periodic or finite (depending on experiment),
- spatial operators: finite differences for ∇ and ∇^2 .

Time evolution is performed using explicit iteration.

C.2 Initialization

Initial conditions are randomly generated or structured:

- X_0 : random or localized distributions,
- Φ_0 : random phase field,
- K_0 : low-amplitude uniform or noise-based field.

Multiple seeds are used to ensure robustness.

C.3 Simulation Loop

Each simulation step includes:

1. update of X using diffusion, damping, and nonlinear terms,
2. update of Φ via diffusion and vorticity coupling,
3. computation of derived fields J , W , g_{eff} ,
4. update of kinetic field K with feedback,
5. normalization (if applicable).

C.4 Measurement Procedure

At each iteration or at convergence:

- sector-specific metrics are computed,
- invariance scores are evaluated,
- continuity residuals are measured,
- boundary and radiation indicators are extracted.

C.5 Statistical Validation

Each configuration is tested over:

- multiple random seeds,
- different parameter sets,
- varying system sizes.

Results are aggregated using mean, minimum, and success rates.

C.6 Stability Criteria

A configuration is considered valid if:

- dynamics remain bounded,
- structures persist over time,
- sector scores exceed defined thresholds.

C.7 Reproducibility Conditions

Reproducibility is ensured by:

- fixed random seeds,
- explicit parameter reporting,
- deterministic update rules.

D Extended Derivations

This appendix provides additional derivations supporting the emergence of physical sectors.

D.1 Flux Structure from Phase Dynamics

Starting from:

$$\Phi_{t+1} = \Phi_t + \alpha \nabla^2 \Phi_t + \beta W_t,$$

we obtain:

$$J = |X| \nabla \Phi,$$

which defines structured propagation pathways.

D.2 Vorticity and Spinor Analogy

The vorticity:

$$W = \nabla \times J$$

naturally generates rotational structures.

These structures exhibit:

- two-component behavior,
- chirality,
- stability under rotation,

which supports the analogy with spinor fields.

D.3 Emergent Geometry

The effective metric:

$$g_{\text{eff}} = 1 + a\rho + b|\nabla^2 X| + c|W| + d|J| + eK$$

can be interpreted as a scalar curvature proxy.

Spatial variation of g_{eff} induces:

- geodesic-like paths,
- effective gravitational behavior,
- curvature-dependent propagation.

D.4 Boundary Radiation Mechanism

Radiation emerges at regions where:

- g_{eff} is high,
- flux gradients are strong,
- kinetic imbalance exists.

This produces:

$$R = R(H, J, W, K, \nabla X, \nabla^2 X),$$

which captures boundary emission.

D.5 Generalized Continuity Origin

The non-zero source term:

$$S_{\text{total}}$$

arises from:

- global normalization constraints,
- kinetic feedback,
- boundary effects.

Thus, conservation laws are effective rather than fundamental.